D-TNet: Category-Awareness Based Difference-Threshold Alternative Learning Network for Remote Sensing Image Change Detection

Ling Wan, Ye Tian, Wenchao Kang, and Lei Ma

***Abstract*— Deep-learning-based change detection methods have achieved remarkable success through the feature learning capability of deep convolutions. However, the network structures of existing methods are simply modified from the semantic segmentation models, ignoring the essential characteristics of change detection, thereby limiting their applications. In this work, we propose a category-awareness-based difference-threshold alternative-learning network (D-TNet) for remote sensing image change detection. Our motivation is to characterize the different change magnitudes for different land cover changes, and represent the semantic content differences of various objects. Thus, our D-TNet consists of a difference map (DM) learning path and a threshold map (TM) learning path, realizing self-adapting threshold selection by assigning each pixel a unique threshold. The two paths are alternatively optimized to make the DM more discriminative, as well as making the TM more adaptive. In addition, a category-awareness attention mechanism is introduced in D-TNet, which learns a pixel-to-category relationship to benefit in representing the heterogeneity of land covers. Finally, experimental results on three change detection datasets verify the effectiveness of our D-TNet in both visual and quantitative analyses. Code will be available at: https://www.researchgate.net/profile/Ling-Wan-4.**

***Index Terms*— Category-awareness, change detection, optical remote sensing image, threshold learning.**

I. Introduction

CHANGE detection is a technique to discriminate change information by analyzing two or more images acquired over the same area at different times [1]. With the rapid development of satellite sensor technology, a large number of remote sensing images have formed multiview and high-revisit observations of the Earth's surface, providing important support for many applications, including agricultural investigation, land use monitoring, and disaster emergency management [2].

Manuscript received 30 May 2022; revised 25 July 2022 and 29 August 2022; accepted 24 September 2022. Date of publication 19 October 2022; date of current version 11 November 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62071466 and Grant 61901439, in part by the Fund of National Key Laboratory of Science and Technology on Remote Sensing Information and Imagery Analysis, Beijing Research Institute of Uranium Geology under Grant 6142A010402, and in part by the Guangxi Natural Science Foundation under Grant 2018GXNSFBA281086. *(Corresponding author: Lei Ma.)*

The authors are with the Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China, and also with the University of Chinese Academy of Sciences, Beijing 100039, China (e-mail: wanling15@mails.ucas.ac.cn; lei.ma@ia.ac.cn).

Digital Object Identifier 10.1109/TGRS.2022.3213925

Over the past few decades, many researchers have made efforts on remote sensing image change detection. Traditional change detection methods, such as the arithmetic operation-based methods [3], [4], the image transformation-based methods [5], [6], [7], [8], and the model-based methods [9], [10], are mainly based on the manually designed complicated features to compare the multitemporal images. Thus, they rely on expert domain knowledge, and are difficult to make obvious breakthrough in accuracy.

Recently, since the recognition of the excellent ability of deep learning models in capturing informative image features, deep learning models have been applied to change detection tasks. Generally, change detection can be regarded as a problem of extracting deep semantic information from the multitemporal images, and then classifying it into changed and unchanged regions. Thus, their network structures are usually modified from the semantic segmentation models; for example, the U-shaped encoder–decoder architecture is the most commonly used one with Siamese or two-channel structures. Moreover, to improve the performance of change detection results, researchers usually adopted dense connection [11], multilevel aggregation [12], [13], [14], and attention mechanism [15], [16], [17], [18], [19] to enhance the feature extraction capability.

However, although existing deep-learning-based methods have been proved to be successful for detecting changes, they simply treat the change detection task as a semantic segmentation task, ignoring the essential characteristics of change detection. Thus, several challenges limit their applications.

A. Different Kinds of Land Covers Require Heterogeneous Feature Extractors

Typically, a remote sensing change detection dataset (CDD) contains various ground objects, such as buildings, roads, trees, vehicles, and bridges. Even for the same kind of objects in a scene, it has multiscale and multishape characteristics. For example, the buildings have large, middle, and small sizes, with square, rectangle, and round shapes. Therefore, change detection tasks require to consider the heterogeneity of different kinds of land covers. However, "relation attention" in existing methods is achieved by using global max or average pooling operations to recalibrate features from the channel and spatial dimensions, highlighting the informative features and

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

suppressing the redundant features [20], [21], [22], [23], [24]. These pooling operations regard all pixels equally, ignoring the semantic content differences between regions, resulting in insufficient discrimination of the captured features.

B. Different Land Cover Changes Require Different Change Thresholds

Different land covers exhibit different spatial heterogeneities, and different land cover changes characterize different change magnitudes. Thus, the selection of change threshold is a complicated process. On the one hand, high spatial resolution images are usually acquired at off-nadir angles. Change in viewing angle causes change in geometrical properties of the images, and thus the images cannot be perfectly registered for all pixels. However, different objects have different sensitivities to this misalignment, resulting in different objects exhibiting different spatial heterogeneities in the feature difference maps (DMs). On the other hand, for the class-specific change detection tasks, for example, building change detection tasks, the nonbuilding areas are identified as background and should be assigned large thresholds. Therefore, the change threshold selection requires to be self-adapting, and considers the influence of the land cover types. However, existing methods aim to enhance the separability of changed and unchanged features, and compute the change map using a pixelwise argmax operation on the channel. Therefore, these methods focus on feature extraction, and ignore the effect of thresholds, leading to serious false alarms in the change detection results.

To tackle the aforementioned challenges, in this work, we propose a category-awareness-based difference-threshold alternative-learning network (D-TNet) for remote sensing image change detection. Our intuition is to characterize the different change magnitudes for different land cover changes. Instead of employing a pixelwise argmax operation on the channel to compute the change map, our D-TNet consists of a DM learning path and a threshold map (TM) learning path, realizing self-adapting threshold selection by assigning each pixel a unique threshold. The two paths are alternatively optimized making the DM more discriminative and making the TM more adaptive. In addition, our D-TNet introduces a category-awareness attention (CAA) mechanism that serves both DM and TM paths. By learning the pixel-to-category relation, the feature representation of heterogeneous land covers is enhanced.

The contributions of this article are summarized as follows.

1) A category-awareness-based D-TNet is proposed for remote sensing image change detection; the proposed network consists of a DM learning path and a TM learning path, which assigns each pixel a unique threshold to achieve self-adapting threshold selection.

2) An alternative optimization strategy is introduced to train D-TNet. The DM and the TM are learned alternatively, making the DM more discriminative and making the TM more adaptive.

3) A CAA mechanism is proposed to learn the pixel-to-category relation, which benefits in representing the heterogeneity of land covers.

4) Extensive experiments on three CDDs verify the effectiveness of the D-TNet. Experimental results show that our method achieves more satisfactory performance in both visual and quantitative analyses.

II. Related Works

A. Traditional Change Detection Methods

The change detection technology has been long investigated. Generally, traditional image processing methods can be roughly divided into three categories, including DM-based methods, classification-based methods, and other advanced model-based methods.

The DM-based methods first generate difference image, and then distinguish the changed and unchanged regions by thresholding or clustering process [3], [4], [5], [6], [7], [8]. The difference image is commonly obtained by image algebra operations [e.g., differencing, ratioing [4], and change vector analysis (CVA) [3]] and image transformation operations [e.g., principal component analysis (PCA) [5], multivariate alteration detection (MAD) [6], subspace learning [7], and sparse learning [8]].

The classification-based methods first independently classify two images, and then regard the pixels or regions falling into different classes as changes [25], [26]. These methods require prior information on land covers or changes, and are able to obtain from–to change types. However, the effectiveness of these methods is highly dependent on the performance of the classifiers, and thus, many machine-learning-based classifiers, such as support vector machine (SVM) [27] and decision tree [28], are employed.

With the advancements in image processing and computer vision, some advanced models are also used for change detection, including extreme learning machine (ELM) [9] and Markov random field [10]. However, the aforementioned methods are based on traditional manually designed features, which are difficult to adapt to numerous change detection scenarios.

B. Deep-Learning-Based Change Detection

In recent years, deep learning algorithms, aiming to automatically learn representative and discriminatory features from a set of data, have received much attention in the change detection field. Deep-learning-based change detection methods can be roughly divided into three categories: unsupervised approaches, self-supervised approaches, and supervised approaches.

1) Unsupervised Methods: The unsupervised methods take advantages of the available pretrained convolutional neural network (CNN) architectures. They first use CNN to exploit multiresolution deep feature maps, and then select the most discriminative features through feature selection techniques. Finally, they identify the changed pixels by analyzing the comparable pixelwise features [29], [30]. However, on the one hand, the performance of these methods is heavily dependent on the feature selection strategy. On the other hand, although deep features exhibit stronger capabilities than shallow features, they ignore the feedback advantages of deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

2) Self-Supervised Methods: The self-supervised methods first generate pseudolabels based on the traditional change detection methods, and then use the pseudolabels to train the CNN [29], [31], [32]. Through the iterative optimization of pseudolabels and CNN, the final change detection result is obtained. However, these methods are sensitive to the pseudolabels. On the one hand, the false pseudolabels directly increase the error rate of change detection results. On the other hand, the pseudolabels have limitations and one-sidedness, making the model to have enough confidence to predict the easier classes which usually dominate self-training.

3) Supervised Methods: With the ever-increasing availability of Earth observation data, more and more CDDs are constructed [15], [16], [33], [34], making the supervised methods to receive much attention and gradually develop. Zhan et al. [35] introduced CNN into the change detection tasks, which utilized a Siamese convolutional network to obtain the DM and then employed the threshold segmentation to achieve the change map. Daudt et al. [36] designed three end-to-end change detection architectures based on fully CNNs, and they are fully convolutional early fusion (FC-EF), fully convolutional Siamese-concatenation (FC-Siam-Conc), and fully convolutional Siamese-difference (FC-Siam-Di). Later, other state-of-the-art deep learning models, such as stacked autoencoder (SAE), deep belief network (DBN), recurrent neural networks (RNNs), long short-term memory (LSTM) [37], generative adversarial network (GAN) [38], [39], and Transformer [40], are also devised and applied in change detection tasks. Considering the features of remote sensing images and the characteristics of change detection tasks, researchers improve the performance of change detection results mainly from the following three perspectives: enhancing feature extraction capability (e.g., dense connection, multilevel aggregation, and receptive field expansion), adding attention mechanism, and loss function modification.

Considering the multiscale characteristics of the changed targets, researchers employ dense connection [11], multilevel aggregation [12], [13], [14], and receptive field expansion [41], [42] to enhance feature extraction capability. Peng et al. [11] adopted UNet++ with dense skip connection backbone to learn multiscale and different semantic levels of visual feature representations. Zheng et al. [12] proposed a cross-layer CNN (CLNet), which designes cross-layer blocks (CLBs) to incorporate the multiscale features and multilevel context information. Hou et al. [13] proposed a high-resolution triplet network (HRTNet), which designes a novel dynamic inception module to enrich the multiscale information of the extracted features. Zhang et al. [14] proposed a satellite image change detection network (CDNet) based on Unet++ and differential pyramid (DifUnet++), which adoptes a multiply side-outs fusion strategy to consider the scale diversity of changed regions in remote sensing images. Zhang et al. [42] took advantage of atrous spatial pyramid pooling (ASPP), which uses different size convolutions for receptive field increment. Lei et al. [41] introduced spatial pyramid pooling (SPP) into the backbone network to increase the receptive field by using multiscale blocks.

Attention-based models aim of emphasizing the critical information of images or features. The most commonly used attention mechanisms include channel attention, spatial attention, and mixed-domain attention [15], [16], [17], [18], [19]. For example, Chen and Shi [16] proposed a Siamese-based spatial–temporal attention neural network (STANet), which introduces a basic spatial–temporal attention module (BAM) to exploit the global spatial–temporal relationship and a pyramid spatial–temporal attention module (PAM) to aggregate multiscale attention representation. Zhang et al. [15] designed a deeply supervised image fusion network (DSIFN), which adoptes attention modules to fuse multilevel deep features with image difference features to improve the internal compactness of the changed regions. Chen et al. [17] introduced dual-attentive fully convolutional Siamese networks (DASNets) that use the dual attention mechanism to capture the long-range dependencies of images. Huang et al. [18] constructed a multiple attention Siamese network (MASNet), which employs an attention feature fusion module (AFFM) to consider global and local channel information simultaneously. Ke and Zhang [19] proposed a cross-Siamese CDNet based on hierarchical-split attention (CS-HSNet), which establishes a lightweight cross spatial-channel triplet attention (CSCTA) module to capture cross-dimensional long-range relationship between triplet combinations: channel with height, channel with width, and channel with channel.

The loss function modification aims to solve the problem of the imbalance of positive and negative samples. Change detection usually follows the loss function of semantic segmentation, such as contrastive loss, cross-entropy loss, and dice coefficient loss. But the change detection task usually suffers from extreme label imbalance, with more than 90% of the pixels being labeled as unchanged. Therefore, research often introduces balanced weights into the loss function to alleviate the sample imbalance problem [12], [21], [35]. In addition, Zhang et al. [14] added a modulating factor to cross entropy loss to highlight hard samples. Du et al. [20] employed the intersection over union (IoU) to measure the similarity between the predicted map and ground truth, which allows to treat any scale-changed objects or areas equally. Wei et al. [22] introduced the structure similarity index measure (SSIM) loss that captures the structural information and assigns higher weights to the boundaries.

However, although existing deep-learning-based methods have been proved to be successful for detecting changes, they simply treat the change detection task as the semantic segmentation task, ignoring the essential characteristics of change detection. In this article, we propose a D-TNet, which is effective in characterizing the different change magnitudes for different land cover changes, as well as modeling the pixel-to-category relationships to benefit the feature representation of heterogeneous land covers.

III. Methodology

The overview of the proposed D-TNet architecture is illustrated in Fig. 1. D-TNet is mainly consisted of backbone feature extraction (BFE) module, category-awareness descriptor

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

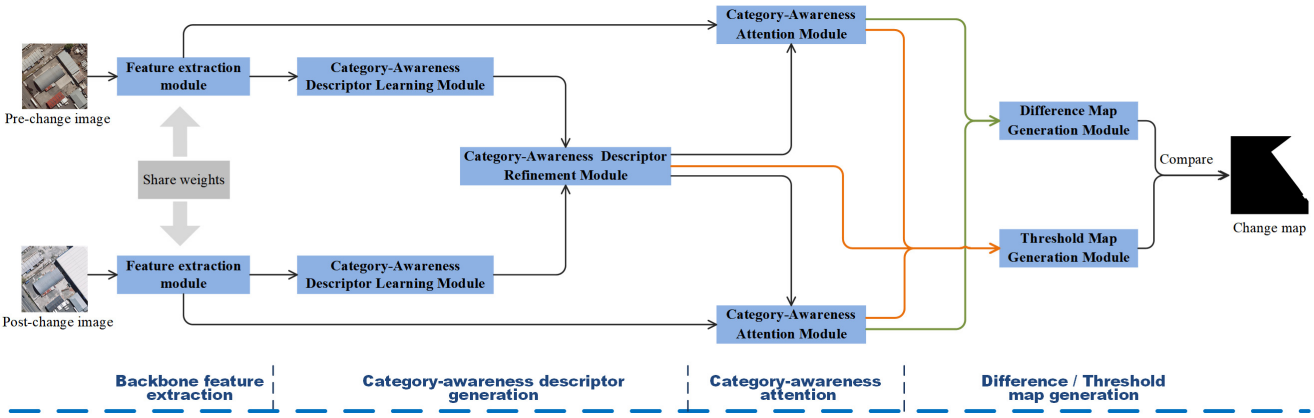


Fig. 1. Overview of the proposed D-TNet architecture.

generation (CADG) module, CAA module, DM generation (DMG) module, and TM generation (TMG) module.

D-TNet starts with the BFE module to capture the feature maps for each temporal image. Then, the feature maps are sequentially fed into the category-awareness descriptor learning (CADL) module and category-awareness descriptor refinement (CADR) module to obtain the enhanced category-awareness descriptors, which subsequently guide the feature map response calibration in the CAA module. BFE, CADG, and CAA modules constitute the feature extraction parts. After that, the category-awareness features and the enhanced category-awareness descriptors are fed into the DMG module and the TMG module. In the DMG module, the DM is calculated by decoding the category-awareness difference features between the two temporal inputs using a lightweight decoder. In the TMG module, the category-awareness features of the two temporal inputs are concatenated, and then guided by the enhanced category-awareness descriptors using pixel-to-category relation, generating the TM. Finally, the DM and the TM are compared to derive the change map.

A. Backbone Feature Extraction

For the backbone network, we choose the Siamese structure, which makes full use of the features of each temporal input. For the basic feature extractor branch, we adopt the ResNet50 with the feature pyramid network (FPN) structure rather than the ResNet50 alone, because the features retrieved by the decoder layers provide more spatial context information about the changes [30].

As shown in Fig. 2, $\{\mathbf{E}_i^t | i = 2, 3, 4, 5\}$ represent the features extracted from the last four stages of ResNet50, which have stride of $\{2^i | i = 2, 3, 4, 5\}$ with respect to the input image. $\{\mathbf{F}_i^t | i = 2, 3, 4, 5\}$ denote the features generated by FPN, implemented through top-down pathway and lateral connections

$$\mathbf{F}_i^t = g\left(\mathbf{E}_i^t\right) + \varphi\left(\mathbf{F}_{i+1}^t\right), \quad i = 2, 3, 4, 5 \tag{1}$$

where g is the lateral connection by 1×1 convolution; φ denotes the nearest neighbor upsampling with a scale factor of 2; t represents the prechange or postchange.

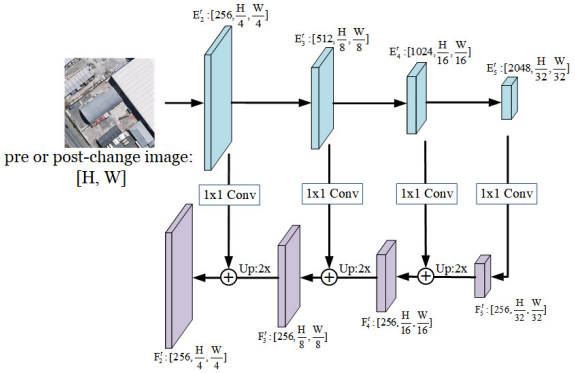


Fig. 2. Illustration of BFE module.

The two branches separately extract hierarchical features from each temporal input. Benefiting from top-down path and lateral connections, features are enhanced by comprehensively utilizing the high spatial information from shallow layers and the strong semantic information from deep layers, which facilitates the subsequent change feature extraction.

B. Category-Awareness Descriptor Generation

A remote sensing image usually contains various kinds of objects, and even the same kind of objects also has multiscale and multishape characteristics. Therefore, an effective change detection method requires to have strong feature extraction ability to represent the heterogeneity of land covers. In other words, it needs to capture more discriminative features for multicategory objects in each scene, while extracting more representative features for multiscale and multishape objects in each category. In order to achieve this goal, we design a CADG module, which consists of two parts: the CADL module and the CADR module. For convenience, in Section III-C, we do not distinguish between multicategory objects in one scene and multiscale and multishape objects in one category, and use "object" uniformly.

1) Category-Awareness Descriptor Learning: To discriminate the semantic content heterogeneity of land covers, we introduce a category-awareness descriptor to provide an overall representation for one category, which is smoother and more global compared with the pixel-level feature.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

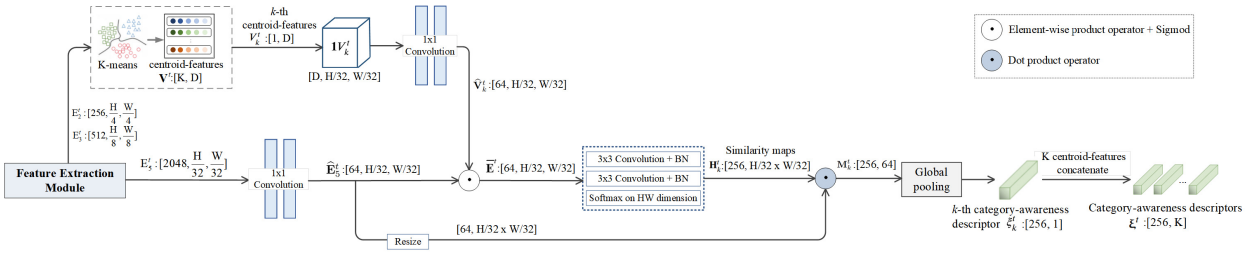


Fig. 3. Illustration of CADL module.

The process of CADL is illustrated in Fig. 3. First, the clustering algorithm is employed to obtain the centroid features of each category. Specifically, for each temporal, we randomly select n images to feed into ResNet50 pretrained on ImageNet to obtain the feature maps. Then, PCA is employed to reduce feature vectors to D dimensions, and K-means is utilized to cluster the D-dimensional feature vectors into K groups, deriving the centroid features. The centroid features are denoted as $\mathbf{V}^t \in \mathbb{R}^{K\times D}$ and the kth centroid feature is represented as ${V_k}^t \in \mathbb{R}^D$, where K is the number of groups; D is the number of feature dimensions. In addition, we need to make one-to-one correspondence between the centroid features of different temporals, obtaining category consistency for the centroid features of multitemporal images.

Based on the centroid features, we extract category-awareness descriptors from the convolutional feature maps. To be more specific, we first expand the kth centroid feature V_k^t into a high-dimensional tensor $\mathbf{1}V_k^t$, and then project $\mathbf{1}V_k^t$ into $\widehat{\mathbf{V}}_k^t$. Then, we encode the convolutional feature maps $\mathbf{E}_5^t$ to $\widehat{\mathbf{E}}_5^t$, and reweight $\widehat{\mathbf{E}}_5^t$ using $\widehat{\mathbf{V}}_k^t$, obtaining $\overline{\mathbf{E}}^t$. After that, we compute the similarity maps $\mathbf{H}_k^t$ through convolution layers and a softmax function operated on the $H\cdot W$ dimension (first reshape, then softmax), which measure the similarity between each pixel and the kth group. Next, we calculate the average sum of pixels in $\overline{\mathbf{E}}^t$ weighted by $\mathbf{H}_k^t$ to save the detailed category-awareness representations $\mathbf{M}_k^t$. The specific formulas are as follows:

$$\begin{cases} \widehat{\mathbf{V}}_k^t = \phi_{\mathbf{W}_V}\left(\mathbf{1}V_k^t\right) \\ \widehat{\mathbf{E}}_5^t = \phi_{\mathbf{W}_E}\left(\mathbf{E}_5^t\right) \\ \overline{\mathbf{E}}^t = \sigma\left(\widehat{\mathbf{E}}_5^t \circ \widehat{\mathbf{V}}_k^t\right) \\ \mathbf{H}_k^t = \text{Softmax}\left[\phi_{\mathbf{W}_\theta}\left(\overline{\mathbf{E}}^t\right)\right] \\ \mathbf{M}_k^t = \mathbf{H}_k^t \cdot T_R\left(\overline{\mathbf{E}}^t\right) \end{cases} \tag{2}$$

where $\phi_{\mathbf{W}_E}(\cdot)$ and $\phi_{\mathbf{W}_V}(\cdot)$ are 1×1 convolutional layers; $\phi_{\mathbf{W}_\theta}$ represents two 3×3 convolutional layers; $\mathbf{W}_\theta, \mathbf{W}_E, \mathbf{W}_V$ are learnable parameters; $\mathbf{1}$ is an all-1 matrix to expand ${V_k}^t$ to $[D, H/32, W/32]$ tensor, and H and W are the height and width of the input image; $\circ$ is the elementwise product operator. σ is the sigmoid function on HW dimension; $T_R(\cdot)$ is a resize operation.

Finally, we aggregate all features in $\mathbf{M}_k^t$ on the HW dimension to obtain the kth category-awareness descriptor ξ_k^t, and the category-awareness descriptors $\boldsymbol{\xi}^t$ are derived by concatenating multigroups ξ_k^t.

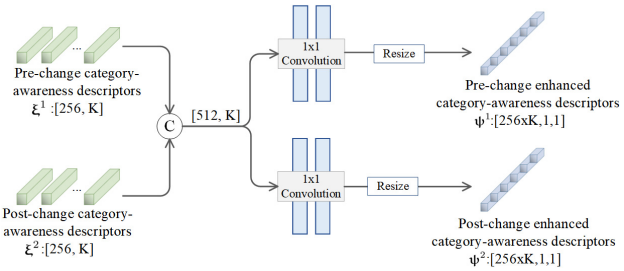


Fig. 4. Illustration of CADR module.

2) Category-Awareness Descriptor Refinement: The category-awareness descriptors generated by the CADL module are independent of each other for different temporals. It is necessary to exploit the space–time relationships to enhance the representation ability of the category-awareness descriptors at each temporal. As illustrated in Fig. 4, the prechange descriptors $\boldsymbol{\xi}^1$ and the postchange descriptors $\boldsymbol{\xi}^2$ are first concatenated, and then fed it into two sets of convolution layers, deriving their enhanced descriptors with a reshape operation. The enhanced category-awareness descriptors are denoted as $\boldsymbol{\psi}^1$ and $\boldsymbol{\psi}^2$.

C. Category-Awareness Attention

Based on the enhanced category-specific descriptors, the CAA module is proposed to improve the discrimination of the feature maps extracted by the BFE module. The CAA module learns a pixel-to-category relation by assigning the category-relevant information to each pixel, aiming to enlarge the intercategory semantic consistency while reducing the intracategory semantic consistency.

Fig. 5 shows the procedure of CAA module. The key of CAA module is to model the pixel-to-category relation. Motivated by [43], first, we align the feature maps $\mathbf{F}_i^t$ and the enhanced category-specific descriptors $\boldsymbol{\psi}^t$ into a shared manifold by learning two projection functions. One function is to transform $\mathbf{F}_i^t$ to $\widehat{\mathbf{F}}_i^t$, and the other is to convert $\boldsymbol{\psi}^t$ to $\widehat{\boldsymbol{\psi}}^t$. Then, we apply an elementwise product operator on $\widehat{\mathbf{F}}_i^t$ and $\widehat{\boldsymbol{\psi}}^t$, getting a hidden feature map $\mathbf{B}_i^t$ [43]. The specific formulas are as follows:

$$\begin{cases} \widehat{\mathbf{F}}_i^t = \phi_{\mathbf{W}_{Fi}}\left(\mathbf{F}_i^t\right) \\ \widehat{\boldsymbol{\psi}}^t = \phi_{\mathbf{W}_\psi}\left(\boldsymbol{\psi}^t\right) \end{cases} \tag{3}$$

$$\mathbf{B}_i^t = \sigma\left(\widehat{\mathbf{F}}_i^t \circ \widehat{\boldsymbol{\psi}}^t\right) \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

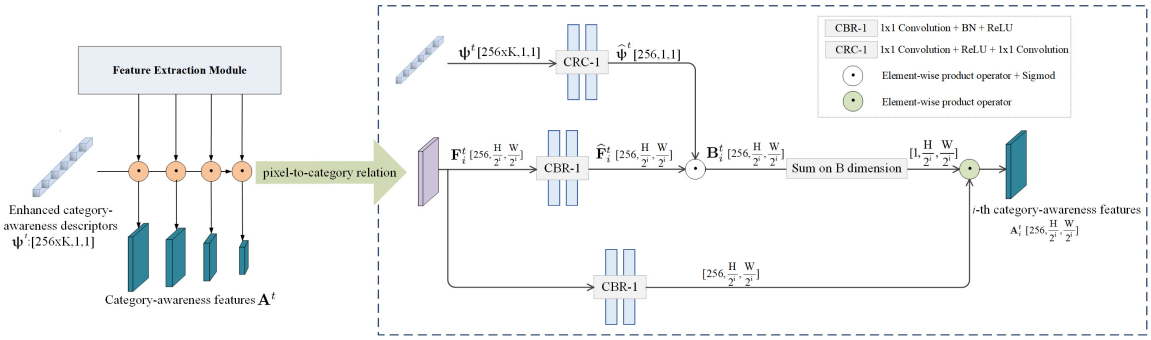


Fig. 5. Illustration of CAA module.

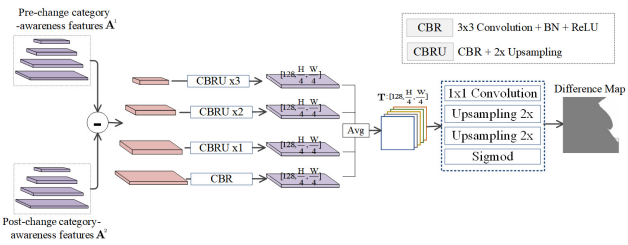


Fig. 6. Illustration of DMG module.

where $\phi_{\mathbf{W}_{Fi}}(\cdot)$ is a 1×1 convolutional layer followed by batch normalization and ReLU; $\phi_{\mathbf{W}_{\psi}}(\cdot)$ is a 1×1 convolutional layer followed by ReLU and a 1×1 convolutional layer; $\mathbf{W}_{Fi}$, $\mathbf{W}_{\psi}$ are learnable parameters; $\circ$ is the pointwise inner product; σ is the sigmoid function.

Next, we calculate the ith category-awareness features $\mathbf{A}_i^t$ by projecting $\mathbf{F}_i^t$, and then reweighting it using $\mathbf{B}_i^t$

$$\mathbf{A}_i^t = \eta\left(\mathbf{B}_i^t\right) \circ \phi_{\mathbf{W}_{\theta i}}\left(\mathbf{F}_i^t\right) \tag{5}$$

where $\phi_{\mathbf{W}_{\theta i}}(\cdot)$ is a 1×1 convolutional layer followed by batch normalization and ReLU; $\eta(\cdot)$ is a sum-up operation on B dimension; $\circ$ is the pointwise inner product.

Finally, apply the aforementioned pixel-to-category relation process to all feature maps $\{\mathbf{F}_i^t | i = 2, 3, 4, 5\}$, and derive the category-awareness features $\{\mathbf{A}_i^t | i = 2, 3, 4, 5\}$, which contain the category-relevant context compared to the original feature maps.

D. DM Generation

Benefiting from the strong feature discrimination and representation capabilities of the category-awareness features, a light decoder is designed to generate DM.

As shown in Fig. 6, given two set of pyramid feature maps $\{\mathbf{A}_i^1 | i = 2, 3, 4, 5\}$ and $\{\mathbf{A}_i^2 | i = 2, 3, 4, 5\}$, We first calculate the difference feature maps in

$$\mathbf{T} = \text{Avg}\{F(\mathbf{X}_2) + \text{UF}(\mathbf{X}_3) + \text{UF}(\text{UF}(\mathbf{X}_4)) + \text{UF}[\text{UF}(\text{UF}(\mathbf{X}_5))]\} \tag{6}$$

where $\mathbf{X}_i = |\mathbf{A}_i^1 - \mathbf{A}_i^2|$; $F(\cdot)$ is a 3×3 convolutional layer (output channel $=128$) followed by batch normalization and ReLU; $\text{UF}(\cdot)$ is the $F(\cdot)$ operation followed by a bilinear upsampling with a scale factor of 2.

Then, a 1×1 convolutional layer, two dense upsampling convolutional layers (scale factor $=2$) [44], and a sigmoid function are sequentially operated on $\mathbf{T}$, to produce the DM with one channel and the same size as the input image.

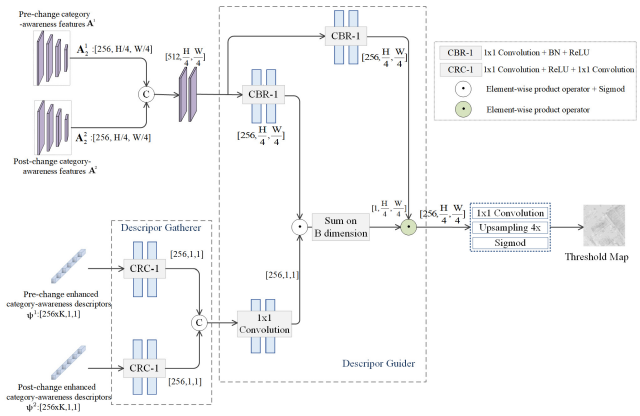


Fig. 7. Illustration of TMG module.

E. TM Generation

Different land covers exhibit different spatial heterogeneities and different sensitivities to imaging conditions, resulting in different land cover changes characterizing different change magnitudes. Therefore, the selection of change threshold is a complicated process, which requires to be self-adapting and considers the influence of the land cover types. However, existing methods usually employ a pixelwise argmax operation on the channel to compute the change map. This method is prone to serious false alarms in the change detection results. To tackle this challenge, we propose a TMG module that assigns each pixel its unique threshold to exploit the influence of change thresholds.

As shown in Fig. 7, we first gather the multitemporal enhanced category-awareness descriptors $\boldsymbol{\psi}^1$ and $\boldsymbol{\psi}^2$ with convolutional layers and concatenation operation. Then, we concatenate the prechange category-awareness features $\mathbf{A}_2^1$ and the postchange category-awareness features $\mathbf{A}_2^2$. Here, we only use $\mathbf{A}^1$ and $\mathbf{A}^2$ with the highest spatial resolution, since the higher the spatial resolution, the more location information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

is preserved, which has a refining effect on the boundaries of change detection results. Next, similar to the CAA module, we use the fusion descriptor to guide the concatenated features, getting the threshold features. Finally, a 1×1 convolutional layer, a nearest neighbor upsampling layer (scale factor $=4$), and a sigmoid function are sequentially operated on threshold features, to produce the TM with one channel and the same size as the input image. Therefore, the TM contains the original image features and the category-relevant context, which is helpful to train the relationship between the DM and the prior information of the input image in a pixel-to-pixel manner.

F. Training Strategy

There are two paths in D-TNet. One is learning the DM, and the other is learning the TM. We train the D-TNet through three phases. First, we freeze the TMG module to train the feature extraction part (BFE, CADG, and CAA modules) and the DMG module. The feature extraction part is the common part of DM generation path and TM generation path. This procedure can be considered as a traditional end-to-end CDNet, which includes high-dimensional feature extraction and DM calculation, and finally generates the change map by argmax operation. Second, we freeze the D-TNet except the TMG module. This procedure is to learn a unique threshold for each pixel based on the DM obtained in phase 1. Third, we freeze the feature extraction part and alternatively optimize the DMG module and TMG module. This procedure is to further improve the change detection performance by enhancing the separability of changed and unchanged features and promoting the rationality of the threshold selection. Finally, we can derive the change map by comparing the DM and the TM. Training details are illustrated in Algorithm 1.

1) DM Learning: To train the DM, we minimize a hybrid loss, which is composed of cross-entropy loss, contrastive loss, and Tversky loss. The formula is as in (7)–(10), shown at the bottom of the page.

L_{ce} is the cross-entropy loss between the DM and the one-hot encoded ground truth. In (8), G_n is the label for the nth pixel; P_n is the change probability in DM for the nth pixels; N is the total number of pixels.

L_{tver} is the Tversky loss based on the Tversky index [45], which controls the tradeoff between recall and precision by hyperparameter α. We set $\alpha = 0.9$ to ensure that the recall of the changes is more important than precision.

Algorithm 1 Training Process of D-TNet

Input: a set of image pairs and corresponding ground truth; parameters include learning rates [phase 1: lr_1; phase 2: lr_2; phase 3: lr_d for training DMG and lr_t for training TMG], batch sizes [phase 1: n_1; phase 2: n_2; phase 3: n_3], training epochs [phase 1: Ep_1; phase 2: Ep_2; phase 3: Ep_3], clustering groups K in Section III-B, feature dimensions D after PCA.

Output: difference maps, threshold maps

(1) pretrain ResNet50 on ImageNet and obtain the centroid-features of each temporal image as described in Section III-B

(2) Phase 1 :

for $Ep \in [1, Ep_1]$ **do**

Freeze the parameters in TMG module, and minimize the objective function in (7) using the back-propagation algorithm to train the feature extraction part and the DMG module.

end for

(3) Phase 2 :

for $Ep \in [1, Ep_2]$ **do**

Fix all parameters in D-TNet except TMG module, and minimize the objective function in (15) using the back-propagation algorithm to learn the threshold map.

end for

(4) Phase 3 :

for $Ep \in [1, Ep_3]$ **do**

1. Fix all parameters in D-TNet except DMG module, and minimize the objective function in (11) to learn the difference map;
2. Fix all parameters in D-TNet except TMG module, and minimize the objective function in (15) to learn the threshold map.

end for

L_{contra} is the contrastive loss. The changed pixels contribute to the loss function only when their change probabilities are within the margin m. We set the hyperparameter $m = 0.5$ to train the change probabilities of changed and unchanged pixels separated from 0.5.

In the third phase, we further train the DMG module by minimizing a modified contrastive loss, which is expressed

$$L_1 = L_{\text{ce}} + L_{\text{tver}} + L_{\text{contra}} \tag{7}$$

$$L_{\text{ce}} = -\frac{1}{N}\sum_{n=1}^{N}\left[G_n \log P_n + (1 - G_n)\log(1 - P_n)\right] \tag{8}$$

$$L_{\text{tver}} = 1 - \frac{\sum_{n=1}^{N} G_n P_n}{\sum_{n=1}^{N} G_n P_n + \alpha \sum_{n=1}^{N} G_n(1 - P_n) + (1 - \alpha)\sum_{n=1}^{N}(1 - G_n)P_n} \tag{9}$$

$$L_{\text{contra}} = \sum_{n=1}^{N}\left[\frac{1}{2}(1 - G_n)(P_n)^2 + \frac{1}{2}G_n\left\{\max(0, m - P_n)^2\right\}\right] \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

as follows:

$$L_2 = \sum_{n=1}^{N}\Bigg[\frac{1}{N_{\text{neg}}}\cdot\frac{1}{2}(1-G_n)(P_n)^2 + \frac{1}{N_{\text{pos}}} \cdot\frac{1}{2}G_n\left\{\max\left(0, PT_n - P_n\right)^2\right\}\Bigg]. \quad (11)$$

In (11), the DM is trained based on the TM. Specifically, a changed pixel contributes to the loss function only when its change probability P_n is smaller than its corresponding change threshold PT_n. In addition, N_{neg} and N_{pos} are the total number of unchanged and changed pixels in ground truth, respectively, to mitigate the problem of category imbalance.

2) TM Learning: To train the TM, we design a loss function based on three principles. First, for the changed pixel, we expect DM to be larger than TM, and the smaller the TM, the smaller the loss value. Second, for the unchanged pixel, we expect DM to be smaller than TM, and the larger the TM, the smaller the loss value. Third, TM should be close to 0.5 to prevent TM from being equal to 0 for all changed pixels and 1 for all unchanged pixels. Therefore, we design the loss function as follows:

$$O_{1n} = \frac{P_n - \text{PT}_n + 1}{2}, \quad O_{2n} = \frac{\text{PT}_n - P_n + 1}{2} \quad (12)$$

$$L_{\text{th}} = -\frac{1}{N}\sum_{n=1}^{N}\left[G_n \cdot \log O_{1n} + (1-G_n)\cdot\log O_{2n}\right] \quad (13)$$

$$L_{\text{norm}} = \frac{1}{N}\sum_{n=1}^{N}(\text{PT}_n - 0.5)^2 \quad (14)$$

$$L_3 = L_{\text{th}} + L_{\text{norm}} \quad (15)$$

where P_n is the change probability for the nth pixel; PT_n is the change threshold for the nth pixel. $P_n, \text{PT}_n \in [0, 1]$, and thus $O_{1n}, O_{2n} \in [0, 1]$.

IV. Experiments

A. Experimental Setup

1) Datasets: We carried out experiments on three CDDs.

a) LEarning, Vision, and remote sensing change detection (LEVIR-CD): The LEVIR-CD [16] is a building CDD (BCDD), which focuses on small and dense building change. It contains 637 image pairs with 0.5 m resolution and 1024 × 1024 size. These images were collected from Google Earth in several cities in Texas, US, ranging from 2002 to 2018. We follow its default dataset split (training/validation/test), and cut the images into small patches of size 512 × 512 without overlap. Thus, we derive 1780/256/512 pairs of patches for training/validation/test.

b) BCDD: The BCDD [34] serves for building change detection and focuses on large-and-sparse building changes. It is acquired by aerial images, and contains two images with the size of 32507 × 15354. It covers an area where a 6.3-magnitude earthquake occurred in February 2011 and rebuilt in the following years. The image acquired in 2012 has 12 796 buildings, and the image acquired in 2016 has 16 077 buildings. We process these datasets into patches of size 512 × 512 without overlap, and randomly divided them into three parts: 924/318/585 for training/validation/test.

c) CDD: The CDD [33] contains 16 000 pairs of images with the size of 256 × 256. These images were taken by Google Earth with a very high spatial resolution of 3–100 cm. The change information in CDD includes buildings, roads, forests, cars, tanks, etc. We follow its default dataset split, and derive 10 000/3000/3000 pairs of patches for training/validation/test.

2) Evaluation Metrics: The quantitative analysis adopts five evaluation metrics to demonstrate the performance of different methods. They are recall, precision, F1-score, IoU of the change category, and overall accuracy (OA). These metrics are defined as follows:

$$\begin{aligned}\text{precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}}\\ \text{recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}}\\ F1 &= \frac{2\cdot\text{precision}\cdot\text{recall}}{\text{precision}+\text{recall}}\\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}}\\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}}\end{aligned} \quad (16)$$

where TP, FP, and FN are the number of true positive, false positive, and false negative, respectively.

3) Implementation Details: All the networks are trained on PyTorch. The experiments were conducted on a machine with an i7-5930K CPU at 3.5 GHz and 32-GB RAM, and an NVIDIA GTX 1080Ti with 10-GB memory. We employ data augmentation to the training data, including flipped horizontally and vertically. We use stochastic gradient descent (SGD) with momentum to optimize our model. The momentum is set to 0.99, and the weight decay is set to 0.0005. The learning rate is initially set to $lr_1 = 0.01$, $lr_2 = 0.001$, $lr_d = 1e^{-4}$, $lr_t = 1e^{-5}$, and we adopt the poly policy to gradually reduce the learning rate, i.e., the learning rate is multiplied by $(1 - (\text{epoch}/\text{maxepoch}))^{\text{power}}$ with $\text{maxepoch} = Ep_n + 50$, power = 2. The minibatch size can reach $n_1 = 4$, $n_2 = 16$, $n_3 = 16$ for LEVIR-CD and BCDD, and $n_1 = 16$, $n_2 = 64$, $n_3 = 64$ for CDD. The training epochs are set to $Ep_1 = 100$, $\text{Ep}_2 = 50$, and $Ep_3 = 50$. Considering the memory capacity, the randomly selected sample number n in the clustering process is set to 20 for LEVIR-CD and BCDD, and 80 for CDD. In addition, the clustering groups K is set to 8, and the feature dimensions D is set to 3. Validation is conducted after each training epoch, and the best model on the validation set is used for evaluation on the test set.

B. Comparison to State-of-the-Art

We make comparisons to several methods, including the purely convolutional-based methods: FC-EF [36], FC-Siam-Di [36], and fully convolutional-Siam-concatenation (FC-Siam-Conc) [36], and related state-of-the-art research in recent years: the deep metric learning CDNet [46], the combination of Siamese network and NestedUNet (SNUNet/32) [21], and the bitemporal image transformer-based model (BIT S4) [40].

FC-EF [36]: An image-level fusion method that concatenates bitemporal images as a single input to the fully convolutional network (FCN).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

FC-Siam-Di [36]: A feature-level fusion method that employs a Siamese FCN to extract each temporal feature and uses difference operation to fuse the bitemporal features.

FC-Siam-Conc [36]: A feature-level fusion method that employs a Siamese FCN to extract each temporal feature and uses concatenation operation to fuse the bitemporal features.

CDNet [46]: A deep metric-learning-based method that employs stacked attention module to enhance the useful information in multiscale features.

SNUNet/32 [21]: A densely connected Siamese FCN-based method that uses compact information transmission between encoder and decoder to mitigate the localization information loss, and uses an ensemble channel attention module to refine different semantic-level features.

BIT S4 [40]: A transformer-based method that expresses the bitemporal image into a few tokens and models contexts in the compact token-based space–time.

For the comparison methods, we use their public codes with default hyperparameters.

1) Qualitative Analysis: Fig. 8 provides the visualization comparison of different methods. For a more intuitive comparison of the results, we use white for true positive, black for true negative, red for false positive, and green for false negative.

For the LEVIR-CD dataset, compared with the competing methods, D-TNet shows the best visual performance with less false detections and misdetections. Specifically, comprehensively observing Fig. 8(b) and (c), for building changes of different scales, we can see that the fully convolutional-based methods, including FC-EF, FC-Siam-Di, and FC-Siam-Conc, suffer from severe false detections and misdetections compared with other methods. This is because the feature extraction ability of fully convolutional-based methods is limited, which is insufficient to distinguish the change information. CDNet, SNUNet, BIT S4, and our D-TNet obtain better results, because they employ attention mechanism and multiscale analysis to enhance the feature extraction capability. As shown in Fig. 8(a) and (c), we can notice that our D-TNet is robust to the pseudochanges caused by different atmospheric conditions, since it obtains the best detection results with the fewest false detections among the seven methods. Observing Fig. 8(d), benefiting from the combination of semantic information, BIT S4 and our D-TNet are robust to the pseudochanges induced by building shadows. However, in comparison, our D-TNet contains less false detections than BIT S4. In addition, the results of our D-TNet have clear building boundaries, because the TMG module uses the features with the highest spatial resolution, which has a refining effect on the boundaries of detection results.

For the BCDD dataset, we observe that our D-TNet achieves the most satisfactory results with high internal compactness and fewer false detections. For example, as shown in Fig. 8(e), the multitemporal images contain changes in seasonal and atmospheric conditions, as well as nonbuilding changes, making it difficult to accurately detect building changes. Our D-TNet obtains the best result with the fewest false detections and misdetections, while other comparison methods suffer from error detections in the building regions. Thus, our model is robust to the appearance differences caused by different imaging conditions. Observing Fig. 8(f) and (g), due to the high resolution of the images and the large size of the city buildings, the texture of the roof is relatively clear. So, for the unchanged buildings, the detection results are likely to contain salt-and-pepper noises, while for the changed buildings, the results are prone to occur incomplete building structures. However, compared with other six methods, D-TNet can overcome this problem thanks to the self-adapting threshold and the category information awareness. As shown in Fig. 8(h), there is an example where no change has taken place between the multitemporal images, and our D-TNet successfully overcomes the interference of pseudochanges. In addition, jointly observing the experimental results of LEVIR-CD and BCDD, our D-TNet can detect the building changes related to urban and nonurban areas, indicating that the proposed method is quite generalized.

For the CDD dataset, different from the LEVIR-CD and BCDD datasets that only detect the building changes, it also monitors the changes caused by cars, trees, roads, etc. Our D-TNet achieves better results than others. For example, as shown in Fig. 8(k), there are changes of very small size cars between multitemporal images. All methods suffer from misdetections, but our D-TNet has the fewest missing detections by comparison, reflecting that our approach pays attention to the multiscale characteristics of remote sensing images and is capable of detecting both small and large changes. As shown in Fig. 8(l), D-TNet successfully detects the changes from trees to green space located at the top right of the image, which benefits from the CAA mechanism. In addition, jointly observing Fig. 8(i)–(l), the results demonstrate that the proposed model can effectively deal with the pseudochanges caused by seasonal differences, and has strong ability to counter against noise. That is, the change detection results of D-TNet have high internal compactness and fewer salt-and-pepper noises.

2) Quantitative Analysis: Table I reports the quantitative analysis on three test sets. The quantitative results show that our D-TNet always outperforms the other methods, which is consistent with the visual inspection. Our D-TNet achieves the highest F1 and IoU values, which means that it obtains a balance between false alarms and misdetections.

3) Model Efficiency: Table I reports the number of parameters (Params.), floating-point operations per second (FLOPs), and the average inference time (ms) for each image in the testing phase. All the experiments are implemented on a computing server equipped with an Intel Core i7-5930K CPU and an NVIDIA GTX 1080Ti GPU. Here, we provide a relatively lightweight version of D-TNet, named D-TNet18, by using Resnet18 instead of Resnet50 as the backbone feature extractor. Similarly, the features extracted from the last four stages of ResNet18 are selected as the original hierarchical features.

As shown in Table I, our D-TNet has the most parameters compared with the other six methods, because it takes Resnet50 as the backbone feature extractor and uses

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

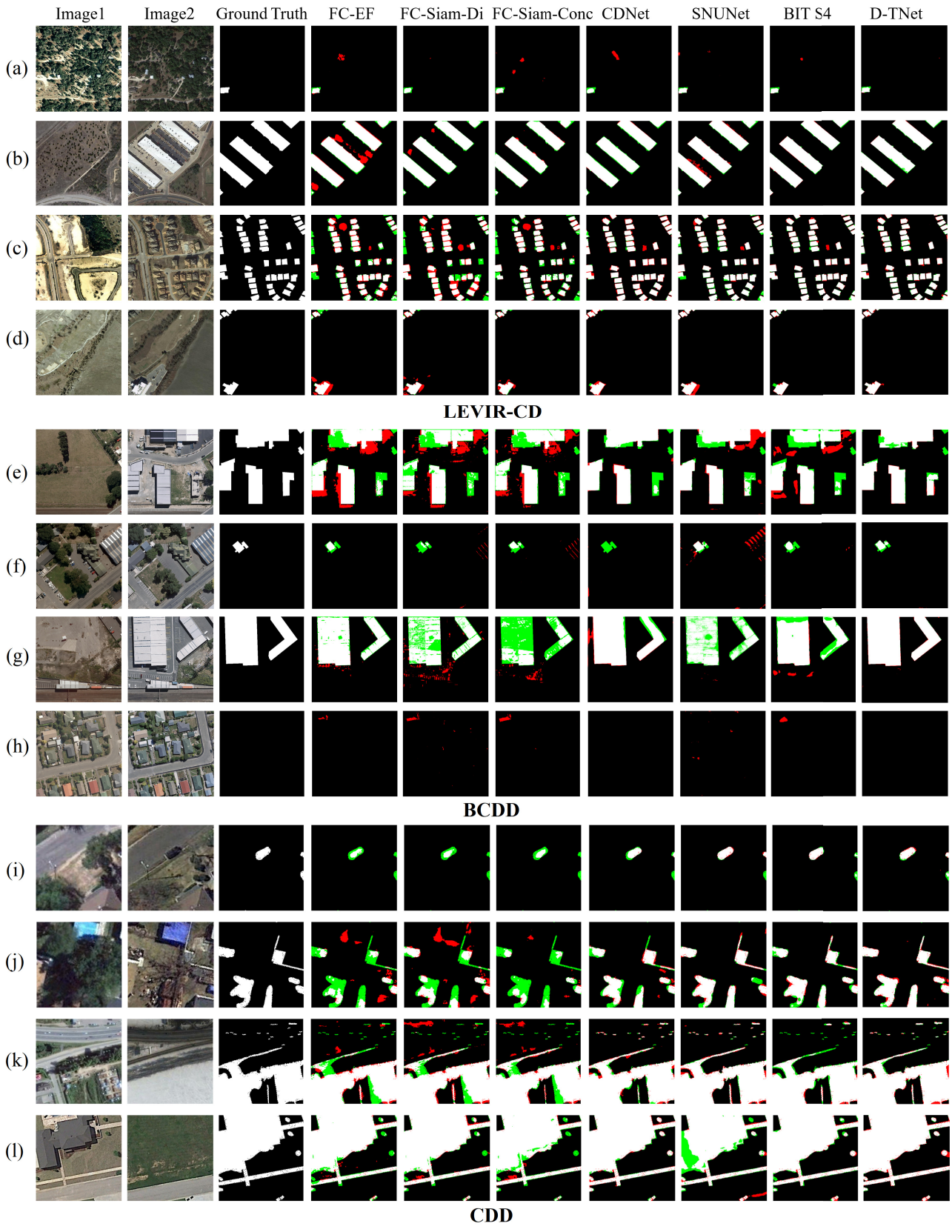


Fig. 8. Change detection results on three datasets. Legend: white for true positive, black for true negative, red for false positive, and green for false negative.

one CADG module and one CAA module for each input image. However, the FLOPs and the inference efficiency of D-TNet are at a moderate level. Combined with the change detection performance, the extra consumption of FLOPs can be compensated for by the improvements in F1-score, IoU, and OA.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE ANALYSIS ON THREE DATASETS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD							
Method	Params.(M)	FLOPs(G)	Time(ms)	recall	precision	F1	IoU	OA
FC-EF	1.29	14.29	∼42	79.85	72.20	75.83	61.07	97.41
FC-Siam-Di	1.29	18.87	∼45	80.99	77.83	79.38	65.81	97.86
FC-Siam-Conc	1.47	21.29	∼45	80.54	80.42	80.48	67.33	98.01
CDNet	16.19	286.32	∼84	91.33	81.52	86.15	75.67	98.50
SNUNet/32	11.49	219.27	∼119	90.97	86.99	88.93	80.07	98.85
BIT_S4	3.39	34.85	∼58	89.37	89.24	89.31	80.68	98.92
D-TNet18	22.75	65.48	∼65	88.58	89.13	88.85	79.94	98.87
D-TNet	35.40	92.49	∼76	90.62	88.75	89.67	81.28	98.94
	BCDD							
Method	Params.(M)	FLOPs(G)	Time(ms)	recall	precision	F1	IoU	OA
FC-EF	1.29	14.29	∼25	71.28	68.09	69.65	53.43	97.76
FC-Siam-Di	1.29	18.87	∼29	65.61	57.54	61.31	44.21	97.01
FC-Siam-Conc	1.47	21.29	∼28	62.71	67.72	65.11	48.28	97.57
CDNet	16.19	286.32	∼67	82.53	87.10	84.75	73.54	98.87
SNUNet/32	11.49	219.27	∼99	84.71	80.02	82.30	69.92	98.46
BIT_S4	3.39	34.85	∼40	81.91	86.15	83.97	72.38	98.62
D-TNet18	22.75	65.48	∼46	85.17	82.99	84.07	72.52	98.83
D-TNet	35.40	92.49	∼55	87.48	88.92	88.19	78.88	99.15
	CDD							
Method	Params.(M)	FLOPs(G)	Time(ms)	recall	precision	F1	IoU	OA
FC-EF	1.29	3.57	∼11	75.57	85.10	80.06	66.74	95.56
FC-Siam-Di	1.29	4.72	∼13	78.99	86.85	82.73	70.55	96.11
FC-Siam-Conc	1.47	5.32	∼14	75.97	90.83	82.74	70.56	96.26
CDNet	16.19	71.58	∼23	91.22	90.07	90.64	82.89	97.78
SNUNet/32	11.49	54.82	∼38	92.04	94.46	93.23	87.33	98.42
BIT_S4	3.39	8.71	∼28	91.73	95.69	93.66	88.07	98.53
D-TNet18	22.75	16.38	∼28	95.01	94.42	94.71	89.96	98.75
D-TNet	35.40	23.13	∼35	96.75	94.06	95.39	91.18	98.90

TABLE II

ABLATION STUDY ON THREE DATASETS. ABLATIONS ARE PERFORMED ON CAA AND TMG. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

			LEVIR-CD			BCDD			CDD		
	CAA	TMG	recall	precision	F1	recall	precision	F1	recall	precision	F1
1	✗	✗	95.30	78.26	85.94	93.08	65.57	76.94	97.93	89.82	93.70
2	✗	✓	91.09	87.12	89.06	91.74	70.59	79.79	95.90	93.87	94.87
3	✓	✗	94.72	82.28	88.06	90.31	80.92	85.36	97.62	92.43	94.96
4	✓	✓	90.62	88.75	89.67	87.48	88.92	88.19	96.75	94.06	95.39

In addition, comparing D-TNet18 and D-TNet, i.e., D-TNet based on different backbones, we can observe that adding more convolutional layers can improve the F1-score while increasing the computational cost. Specifically, the F1-score is improved by 0.82%, 4.12%, and 0.68% on the LEVIR-CD, BCDD, and CDD datasets, respectively. But the computational cost increases by 6.75 giga FLOPs (GFLOPs) and 12.65 M parameters for $256 \times 256 \times 3$ input images.

Therefore, our D-TNet achieves the best F1-score and OA with acceptable computational costs and inference times. However, D-TNet has a large number of parameters, which is the work to be improved in the future.

C. Ablation Studies

To give a more thorough understanding of the proposed D-TNet, we present ablation studies in this section. The performance of different combinations is shown in Table II.

1) Effect of CAA Mechanism: We perform ablation on the effect of CAA mechanism by replacing the CADL module with average pooling operation. As shown in Table II, comparing the quantitative results of Combination 1 with Combination 3, and Combination 2 with Combination 4, we can see that the CAA mechanism is critical in our D-TNet. Influenced by imaging conditions, severe pseudochange in change detection results is a common problem. The CAA mechanism considers the semantic content differences between regions, enhancing the discrimination of the captured features. Thus, it can greatly reduce the false alarms and effectively improve the change detection performance.

2) Effect of TMG Module: We perform ablation on the effect of TMG module by removing it from the D-TNet. As shown in Table II, comparing the quantitative results of Combination 1 with Combination 2, and Combination 3 with Combination 4, we can observe that the model without TMG

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

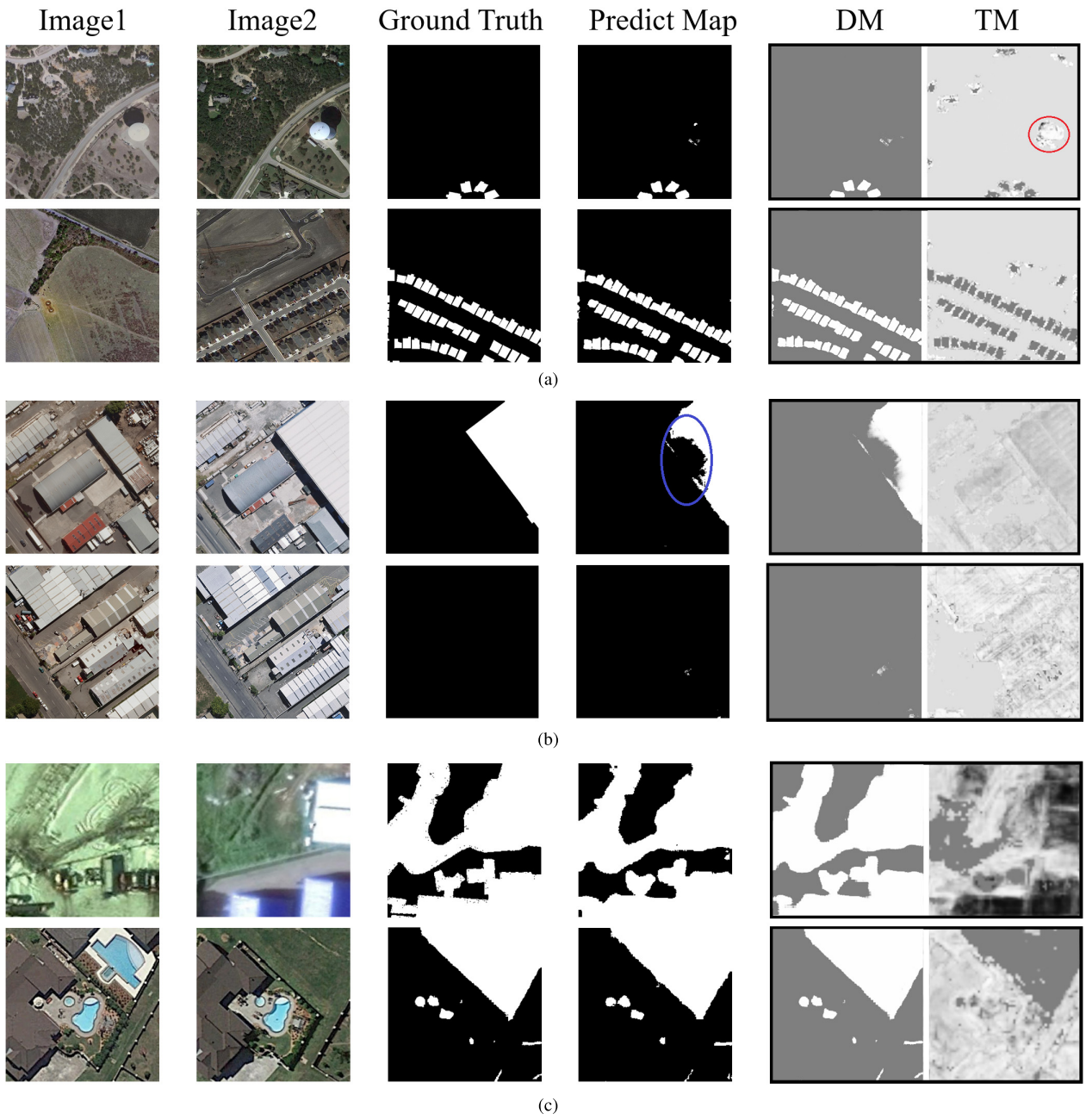


Fig. 9. Effect of TMG module. (a) LEVIR-CD. (b) BCDD. (c) CDD.

module receives significant drops in the values of precision and F1, which proves that the TMG module is critical in our D-TNet.

In addition, Fig. 9 provides some examples of DMs and TMs. If the value of DM is greater than that of the TM, it is determined as changed region; otherwise, it is considered as unchanged region. As shown in Fig. 9, we can see that different regions receive different thresholds, and for changed regions, the TM value is relatively low, which indicates the performance of self-adapting threshold selection. Moreover, TM is related to the category of targets. For example, the TM values of building areas are usually lower. The TM values of boundaries or the spaces between buildings are usually larger, which can effectively reduce false alarms caused by inaccurate alignment in high-resolution images. However, there are some errors in the change detection results. For example, in the area circled in red, the pixel-level TM has the effect of noise, resulting in the detection errors. In the area circled in blue, D-TNet fails to distinguish similar but different buildings. We will solve these issues in our further work.

D. Parameter Analysis

According to Algorithm 1, in addition to the parameters related to network training, including learning rates, batch sizes, and training epochs, there are two parameters in our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PARAMETER ANALYSIS OF FEATURE DIMENSIONS D ON THREE DATASETS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD			BCDD			CDD		
	recall	precision	F1	recall	precision	F1	recall	precision	F1
$D = 1$	90.72	88.48	89.59	87.43	83.16	85.25	93.89	93.87	93.88
$D = 2$	90.59	88.70	89.64	87.12	86.47	86.79	94.13	93.96	94.04
$D = 3$	90.62	88.75	89.67	87.48	88.92	88.19	96.75	94.06	95.39
$D = 4$	90.86	88.50	89.66	87.33	87.57	87.45	95.01	94.33	94.67
$D = 5$	89.77	90.39	90.08	85.24	89.29	87.22	94.69	94.34	94.51
$D = 6$	91.14	88.11	89.60	87.01	84.43	85.70	94.83	93.65	94.24
$D = 7$	90.37	88.84	89.60	88.80	81.05	84.75	95.78	92.41	94.07
$D = 8$	89.92	88.39	89.15	87.58	83.45	85.47	95.12	92.39	93.74

TABLE IV
PARAMETER ANALYSIS OF CLUSTERING GROUPS K ON THREE DATASETS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD			BCDD			CDD		
	recall	precision	F1	recall	precision	F1	recall	precision	F1
$K = 6$	89.32	87.26	88.28	86.92	84.51	85.70	94.82	92.29	93.54
$K = 8$	90.62	88.75	89.67	87.48	88.92	88.19	96.75	94.06	95.39
$K = 10$	89.70	87.46	88.57	87.09	84.93	86.00	95.05	93.08	94.11

TABLE V
DISCUSSION ON TRAINING STRATEGY. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD			BCDD			CDD		
	recall	precision	F1	recall	precision	F1	recall	precision	F1
Phase 1	94.72	82.28	88.06	90.31	80.92	85.36	97.62	92.43	94.96
Phase 2	92.49	85.87	89.06	89.12	84.43	86.72	97.34	93.06	95.15
Phase 3	90.62	88.75	89.67	87.48	88.92	88.19	96.75	94.06	95.39

D-TNet, one is clustering groups K and the other is feature dimensions D.

1) Feature Dimensions D: D serves for the feature dimension reduction before K-means, and is directly represented as the feature dimension of the centroid features. Table III provides the parameter analysis of D on three datasets. We can observe that in general the F1-score rises in the range of D from 1 to 3, then it stabilizes in the range of D from 3 to 5, and then it starts to decline when D is greater than 5. Because the centroid features are used to generate the category-awareness descriptors, whose aim is to make the category-awareness features more discriminative. A small D lacks discrimination, and a larger D increases ambiguity. Thus, it is more appropriate to set D as 3, 4, or 5 according to the experimental results. In addition, the selection of D affects the parameter numbers of the CADL module, i.e., the 1×1 convolutional layer at the top of Fig. 3. Considering the aforementioned ones, we set $D = 3$.

2) Clustering Groups K: Clustering group K is an important parameter in D-TNet, and the clustering results serve for subsequent CAA and TMG modules. Table IV provides the parameter analysis of K on three datasets. We can observe that the quantitative results of $K = 8$ consistently outperform $K = 6$ and $K = 10$. Because K does not represent the exact number of target categories. K should be large enough to capture the heterogeneity of multicategory objects in one scene and multiscale and multishape objects in one category, but not too large to prevent increased ambiguity. Since our goal is binary change detection, if two category objects have similar characteristics, then they can be denoted by one group. Through experimental analysis, 8 is the suitable number of groups, and thus we set $K = 8$.

E. Discussion on Training Strategy

According to Algorithm 1, we train the D-TNet through three phases. Phase 1 is to train the feature extraction part and the DMG module. Phase 2 is to train the TMG module. Phase 3 is to alternatively optimize the DMG module and TMG module.

Table V and Fig. 10 provide the quantitative results and visual results after different phases, respectively. We can observe that compared to Phase 1, Phase 2 can greatly reduce false positives without reducing false negatives, which proves the effectiveness of the TMG module. Phase 3 further balances the false positives and the false negatives through iteratively learning, enhancing the change detection performance. In addition, as shown in Fig. 10, we can see that the boundaries of the detected changed regions are more accurate. This is because the input of the TMG module is the low-level category-awareness features, which add location information to the DM and refine the boundaries of the change detection results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

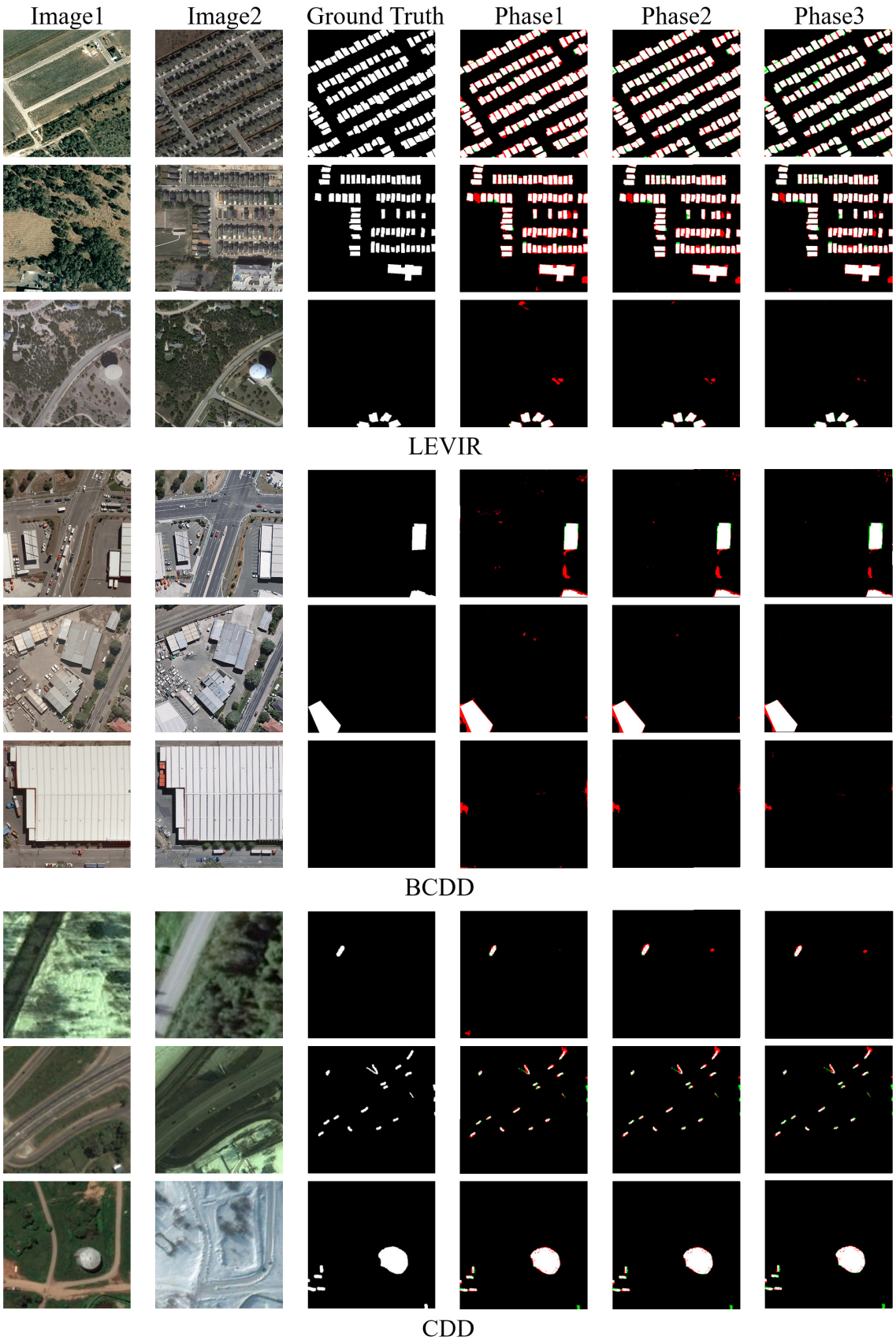


Fig. 10. Change detection results after different phases. Legend: white for true positive, black for true negative, red for false positive, and green for false negative.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this work, we propose a category-awareness-based D-TNet for remote sensing image change detection. Our D-TNet consists of a DM learning path and a TM learning path. On the one hand, D-TNet can characterize the different change magnitudes for different land cover changes by the self-adapting threshold selection. On the other hand, D-TNet can represent the heterogeneous semantic content of various land covers by the CAA mechanism. In addition, the DM and the TM are alternatively optimized to make the DM more discriminative, as well as making the TM more adaptive. Extensive experiments on three CDDs have validated the effectiveness of our D-TNet in both visual and quantitative analyses.

References

[1] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, 1989.

[2] L. Wan, M. Liu, F. Wang, T. Zhang, and H. J. You, "Automatic extraction of flood inundation areas from SAR images: A case study of Jilin, China during the 2017 flood disaster," *Int. J. Remote Sens.*, vol. 40, no. 13, pp. 5050–5077, Jul. 2019.

[3] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[4] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2006.

[5] B. Parmentier, "Characterization of land transitions patterns from multivariate time series using seasonal trend analysis and principal component analysis," *Remote Sens.*, vol. 6, no. 12, pp. 12639–12665, Dec. 2014.

[6] A. A. Nielsen, "The regularized iteratively reweighted mad method for change detection in multi-and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[7] C. Wu, B. Du, and L. Zhang, "A subspace-based change detection method for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 815–830, Apr. 2013.

[8] A. Ertürk, M.-D. Iordache, and A. Plaza, "Sparse unmixing-based change detection for multitemporal hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 708–719, Feb. 2016.

[9] F. Gao, J. Dong, B. Li, Q. Xu, and C. Xie, "Change detection from synthetic aperture radar images based on neighborhood-based ratio and extreme learning machine," *J. Appl. Remote Sens.*, vol. 10, no. 4, 2016, Art. no. 046019.

[10] C. Benedek and T. Szirányi, "Change detection in optical aerial images by a multilayer conditional mixed Markov model," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 10, pp. 3416–3430, Oct. 2009.

[11] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.

[12] Z. Zheng, Y. Wan, Y. Zhang, S. Xiang, D. Peng, and B. Zhang, "CLNet: Cross-layer convolutional neural network for change detection in optical remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 247–267, May 2021.

[13] X. Hou, Y. Bai, Y. Li, C. Shang, and Q. Shen, "High-resolution triplet network with dynamic multiscale feature for change detection on satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 103–115, Jul. 2021.

[14] X. Zhang et al., "DifUnet++: A satellite images change detection network based on UNet++ and differential pyramid," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2021, Art. no. 8006605.

[15] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[16] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[17] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[18] J. Huang, Q. Shen, M. Wang, and M. Yang, "Multiple attention Siamese network for high-resolution image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[19] Q. Ke and P. Zhang, "CS-HSNet: A cross-Siamese change detection network based on hierarchical-split attention," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9987–10002, 2021.

[20] H. Du et al., "Bilateral semantic fusion Siamese network for change detection from multitemporal optical remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2021, Art. no. 6003405.

[21] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[22] H. Wei, R. Chen, C. Yu, H. Yang, and S. An, "BASNet: A boundary-aware Siamese network for accurate remote-sensing change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] Y. Shen et al., "BDANet: Multiscale convolutional neural network with cross-directional attention for building damage assessment from satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5402114.

[24] Y. Lei, D. Peng, P. Zhang, Q. Ke, and H. Li, "Hierarchical paired channel fusion network for street scene change detection," *IEEE Trans. Image Process.*, vol. 30, pp. 55–67, 2021.

[25] M. Volpi, D. Tuia, F. Bovolo, M. Kanevski, and L. Bruzzone, "Supervised change detection in VHR images using contextual information and support vector machines," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 20, pp. 77–85, Feb. 2013.

[26] Y. Li, M. Gong, L. Jiao, L. Li, and R. Stolkin, "Change-detection map learning using matching pursuit," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4712–4723, Aug. 2015.

[27] T. Habib, J. Inglada, G. Mercier, and J. Chanussot, "Support vector reduction in SVM algorithm for abrupt change detection in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 606–610, Jul. 2009.

[28] M. A. Friedl and C. E. Brodley, "Decision tree classification of land cover from remotely sensed data," *Remote Sens. Environ.*, vol. 61, no. 3, pp. 399–409, Sep. 1997.

[29] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[30] L. Bergamasco, S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised change detection using convolutional-autoencoder multiresolution features," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022.

[31] L. T. Luppino et al., "Deep image translation with an affinity-based change prior for unsupervised multimodal change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 4700422.

[32] Q. Li et al., "Unsupervised hyperspectral image change detection via deep learning self-generated credible labels," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9012–9024, 2021.

[33] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, no. 2, pp. 565–571, May 2018.

[34] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[35] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[36] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[37] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral-spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[38] B. Hou, Q. Liu, H. Wang, and Y. Wang, "From W-Net to CDGAN: Bitemporal change detection via deep learning techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1790–1802, Mar. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.

[39] S. Saha, F. Bovolo, and L. Bruzzone, "Building change detection in VHR SAR images via unsupervised deep transcoding," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1917–1929, Mar. 2020.
[40] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.
[41] T. Lei, Y. Zhang, Z. Lv, S. Li, S. Liu, and A. K. Nandi, "Landslide inventory mapping from bitemporal images using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 982–986, Jun. 2019.
[42] C. Zhang, S. Wei, S. Ji, and M. Lu, "Detecting large-scale urban land cover changes from very high resolution remote sensing images using CNN-based classification," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 4, p. 189, Apr. 2019.
[43] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4096–4105.
[44] P. Wang et al., "Understanding convolution for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1451–1460.
[45] A. Tversky, "Features of similarity," *Psychol. Rev.*, vol. 84, no. 4, p. 327, Jul. 1977.
[46] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.



Ling Wan received the B.S. degree from China Agricultural University, Beijing, China, in 2015, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, in 2020.

She is currently an Assistant Researcher with the Institute of Automation, CAS. Her research interests include synthetic aperture radar (SAR) remote sensing image change detection and feature detection.



Ye Tian received the B.S. degree from the China University of Mining and Technology, Beijing, China, in 2012, and the M.E. degree from the China University of Mining and Technology, Beijing, in 2015.

He is currently an Engineer with the Institute of Automation, Chinese Academy of Sciences, Beijing. His research interests include image processing and knowledge map mining analysis.



Wenchao Kang received the B.S. degree from the Beijing Institute of Technology, Beijing, China, in 2016, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, in 2021.

He is currently an Assistant Researcher with the Institute of Automation, CAS. His research interests include optical and synthetic aperture radar (SAR) remote sensing image landcover classification.



Lei Ma received the Ph.D. degree in pattern recognition and intelligent system from the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, China, in 2011.

He is currently an Associate Professor with the Institute of Automation, CAS. His research interests include image processing and remote sensing intelligent interpretation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:01:41 UTC from IEEE Xplore. Restrictions apply.